Zoobenthic Community Indicators of Sediment Contamination

Contents

1 Introduction **1**

2 Contamination Assessment **4**

3 Taxa Assemblage in Reference Sites **6**

3.1 Significant Indicator Taxa 9

4 Environment Driven Taxa Clusters **12**

5 Community Composition Measures **16**

Chapter 1

Introduction

Key Model Parameters

Stage 1: Contamination Assessment

Table 1.1: Stage 1 – Contamination Assessment Parameters

Parameter	**Value**	**Description**
Transform method	log_z_score	Transformation applied to chemical variables
PC1 weight	1.0	Weight for principal component 1
PC2 weight	1.0	Weight for principal component 2
PC3 weight	2.0	Weight for principal component 3 (biologically relevant)
PC4 weight	0.0	Weight for principal component 4
PC5 weight	1.0	Weight for principal component 5
PC6 weight	0.5	Weight for principal component 6
Save individual PCs	True	Save individual PC scores (PC1-PC6)
Standardize PCA scores	True	Standardize PC scores after PCA
Run RDA validation	True	Run RDA validation for pollution
RDA percentile threshold	50	Percentile for RDA reference sites

Stage 2: Taxa Assemblage on Reference Sites

Table 1.2: Stage 2 – Taxa Assemblage Parameters

Parameter	Value	Description
Reference percentile	52%	Bottom X% of sites as reference (least polluted)
Species transformation	hellinger	Transformation method for species data
Number of clusters	3	Number of habitat clusters to identify
Top N taxa	16	Number of top taxa for visualizations
Linkage method	ward	Hierarchical clustering linkage method
Habitat variables	6	Environmental variables for habitat comparison

Habitat Variables (Stage 2)

- Measured Depth (m)
- Velocity at bottom (m/sec)_Imputed
- Water DO Bottom (mg/L)
- Temperature (°C)
- MPS (Phi)
- LOI (%)

Stage 3: Environment-Driven Taxa Clusters (RDA + LDA)

Table 1.3: Stage 3 – RDA and LDA Parameters

Parameter	Value	Description
RDA permutations	999	Number of permutation tests for RDA significance
LDA iterations	1000	Monte Carlo cross-validation iterations for LDA
Taxa transformation	hellinger	Species transformation for RDA
Top N taxa (RDA)	All	Top N taxa for RDA (None = all taxa)
RDA log-transform env	False	Log-transform environmental variables for RDA
RDA standardize env	True	Standardize environmental variables for RDA
LDA standardize env	True	Standardize environmental variables for LDA
LDA test size	0.2	Cross-validation test proportion
Random state	42	Random seed for reproducibility

Stage 4: Community Composition Ordination (PCA-based ZCI)

Table 1.4: Stage 4 – Community Composition Parameters

Parameter	Value	Description
Taxa transformation	hellinger	Species transformation method
Training percentile	100.0%	Percentage of sites to train PCA per cluster
Top N taxa	16	Number of top taxa in loading plots
PC variance threshold	70%	Cumulative variance for PC selection
Min PC variance	5.0%	Minimum variance threshold for regression analysis
Consistent taxa order	True	Use same taxa order across panels
Normalize PCs	False	Z-score normalize PCs for ZCI
ZCI computation method	least_pollution_threshold	Method for ZCI calculation
Least pollution threshold	25%	Bottom X% as reference for ZCI
Ordination PC plane	(1, 2)	Principal components for ordination plot (PC1 vs PC2)

Chapter 2

Contamination Assessment

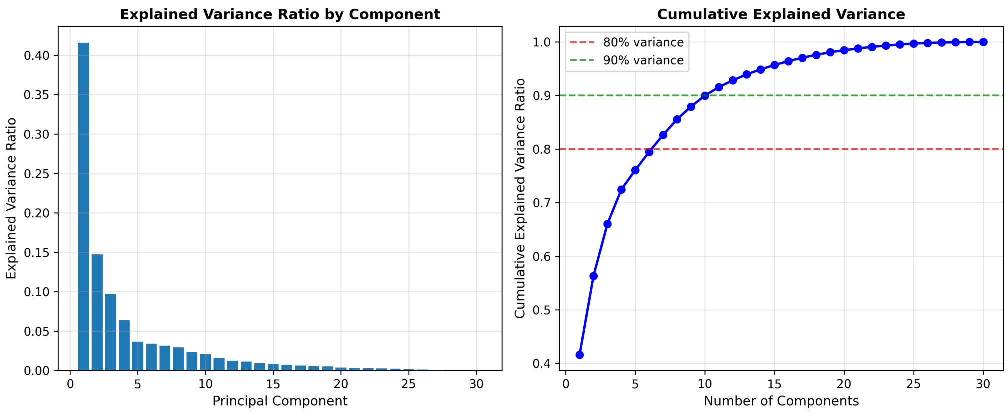

Figure 2.1: PCA Variance

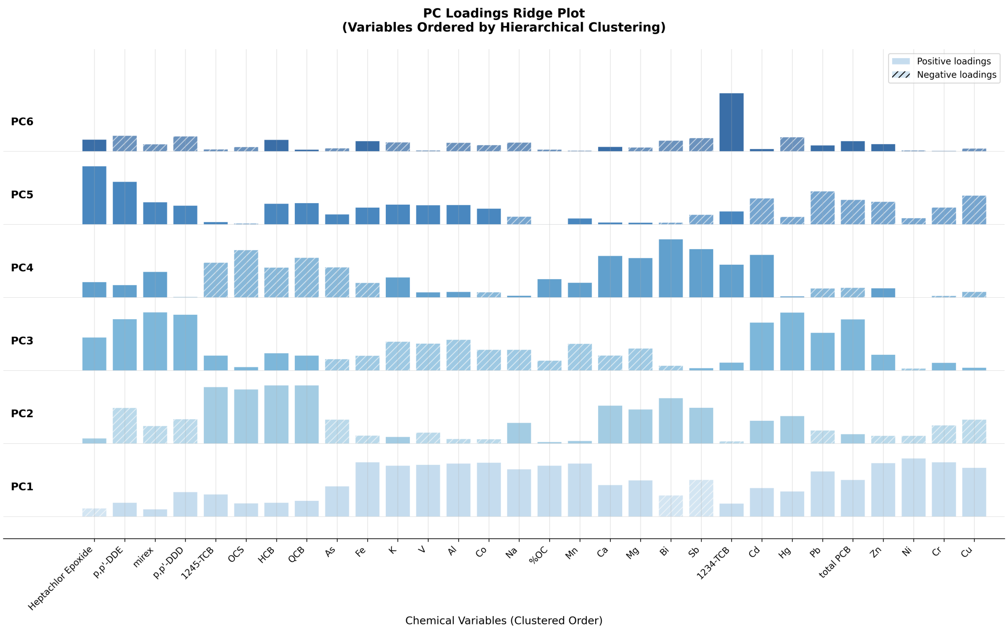

Figure 2.2: PC Loadings Ridge

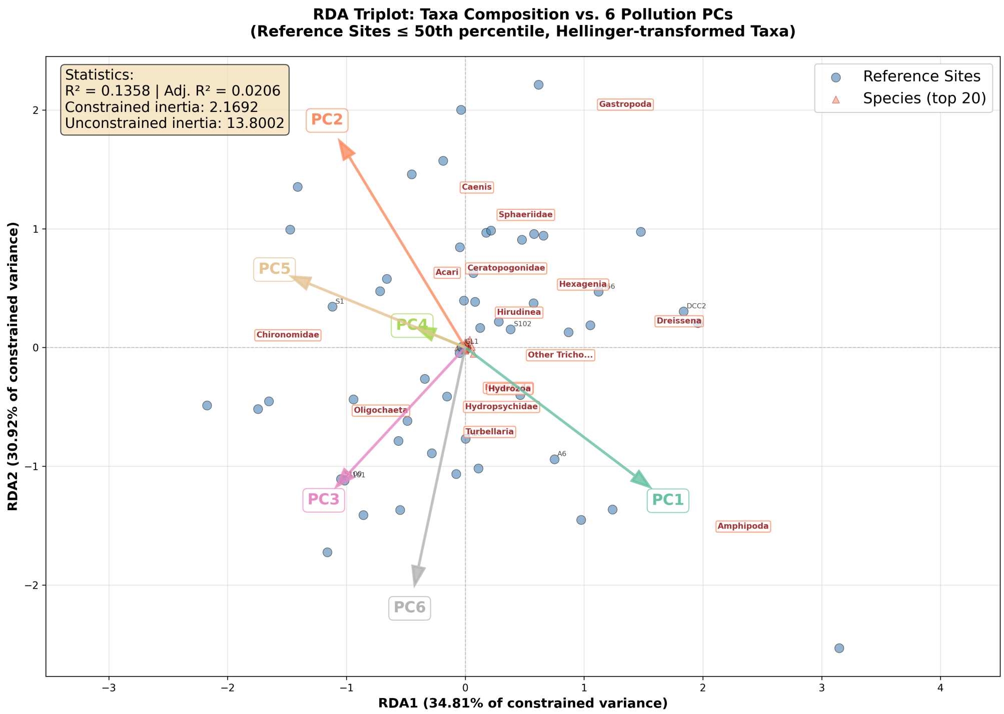

Figure 2.3: RDA Triplot Pollution Species

Chapter 3

Taxa Assemblage in Reference Sites

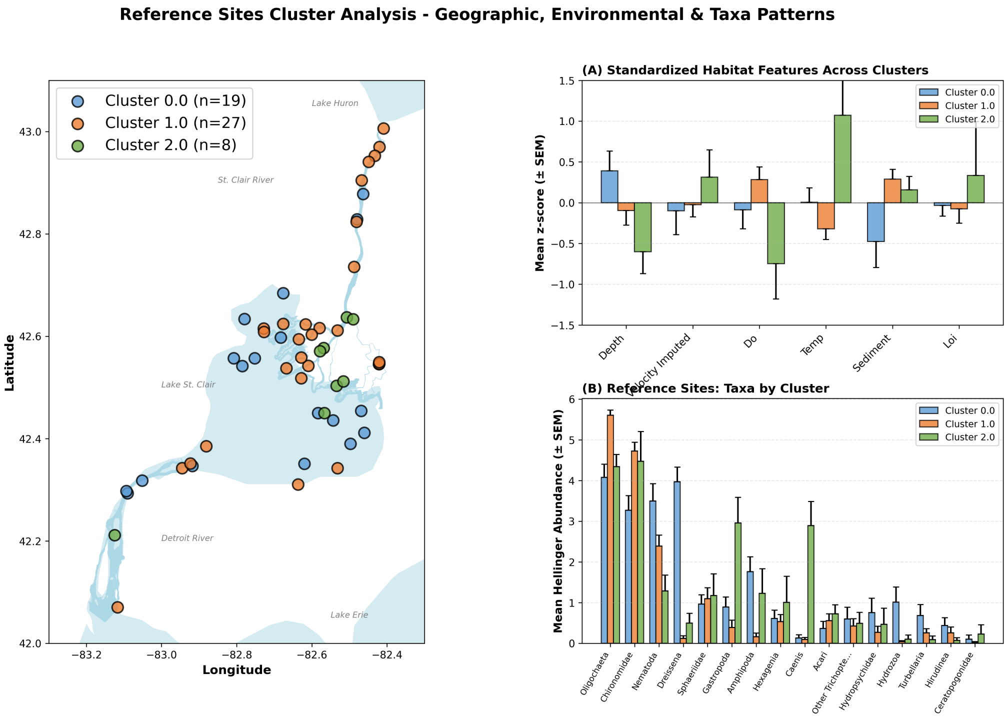

Figure 3.1: Cluster Visualization

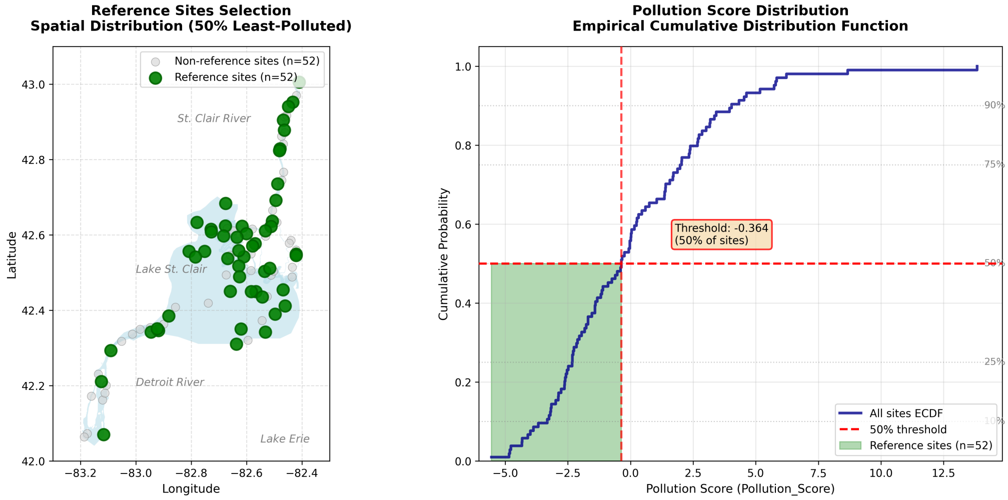

Figure 3.2: Reference Selection

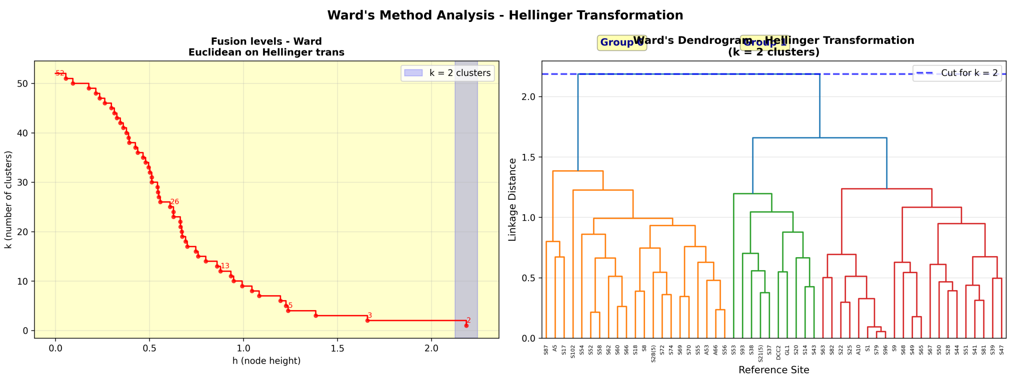

Figure 3.3: Ward Analysis

Table 3.1: Environmental Characteristics by Cluster (Reference Sites Only)

Variable	Cluster 0	Cluster 1	Cluster 2	F-stat
Depth (m)	4.48 ± 2.74	3.50 ± 2.48	2.06 ± 2.04	2.62
Velocity (m/sec)	0.09 ± 0.13	0.12 ± 0.09	0.16 ± 0.11	1.20
DO Bottom (mg/L)	9.44 ± 0.96	9.77 ± 0.73	8.83 ± 1.12	3.68*
Temperature (°C)	20.64 ± 1.38	19.90 ± 1.27	22.47 ± 2.90	7.90**
MPS (Phi)	0.85 ± 1.34	1.50 ± 0.60	1.38 ± 0.46	2.88.
LOI (%)	1.77 ± 0.68	1.75 ± 1.04	2.20 ± 2.14	0.49
Sample size (n)	18	28	8	
Environment Type	Fine Silt	Silt / Organic Rich	Coarse Gravel/Cobble	

Significance: *** p<0.001, ** p<0.01, * p<0.05, . p<0.1

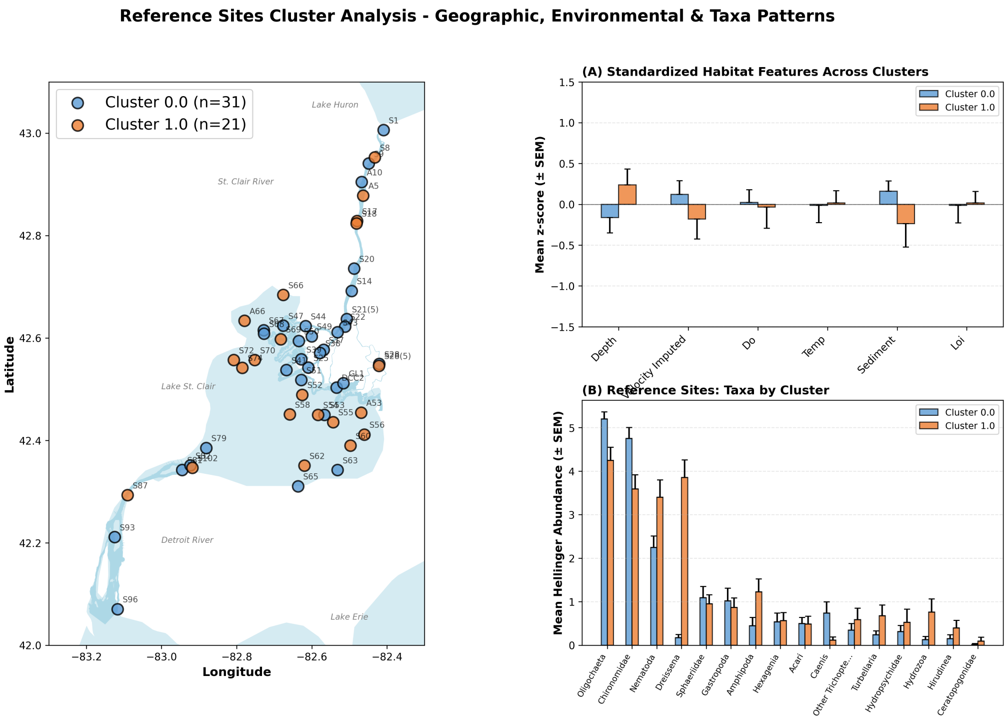

Figure 3.4: Cluster Visualization Final

Table 3.2: Top Taxa Abundance by Cluster (Reference Sites Only)

Taxon	Cluster 0	Cluster 1	Cluster 2	F-stat
Oligochaeta	3.964	5.614	4.337	17.52***
Chironomidae	3.365	4.689	4.471	4.90*
Nematoda	3.478	2.468	1.283	5.71**
Dreissena	3.982	0.114	0.495	86.56***
Sphaeriidae	1.014	1.057	1.169	0.04
Gastropoda	0.940	0.373	2.957	16.36***
Amphipoda	1.728	0.150	1.227	10.27***
Hexagenia	0.604	0.544	1.006	0.59
Acari	0.382	0.566	0.722	0.56
Caenis	0.134	0.088	2.892	57.94***

Showing top 10 of 16 total taxa. Significance: *** p<0.001, ** p<0.01, * p<0.05, . p<0.1

3.1 Significant Indicator Taxa

Seven taxa showed statistically significant differences in abundance among the three reference site clusters (Table 3.2), indicating their potential as habitat indicators.

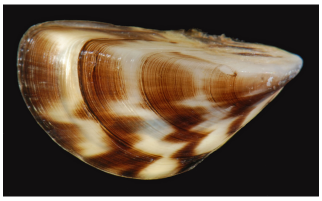

Figure 3.5: Dreissena

Dreissena (F = 86.56***) exhibited the most significant difference among clusters, with extremely high abundance in Cluster 0 (3.982) but nearly absent in Clusters 1 (0.114) and 2 (0.495). Zebra and quagga mussels strongly characterize hard substrate environments with good water circulation. These invasive bivalves are filter feeders that can significantly alter local nutrient dynamics and water clarity. Their presence typically indicates stable, hard substrates such as rocks, cobbles, or artificial structures. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna.

Figure 3.6: Caenis

Caenis (F = 57.94***) was predominantly found in Cluster 2 (2.892) with minimal presence in Clusters 0 (0.134) and 1 (0.088). These small mayflies are characteristic of erosional, shallow-water environments with moderate current. They are collector-gatherers that feed on fine particulate organic matter and are moderately tolerant to pollution, making them useful indicators of intermediate water quality conditions in lotic systems. Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi.

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noindent**Oligochaeta** (F = 17.52***) showed the highest abundance in Cluster 1 (5.614) compared to Cluster 0 (3.964) and Cluster 2 (4.337). These segmented worms are important indicators of organic enrichment in fine-grained sediments. They play a crucial role in sediment bioturbation and nutrient cycling, and their high abundance often indicates depositional environments with accumulated organic matter. Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante.

Figure 3.7: Oligochaeta

Gastropoda (F = 16.36***) showed highest abundance in Cluster 2 (2.957), moderate in Cluster 0 (0.940), and lowest in Cluster 1 (0.373). Freshwater snails prefer hard substrates and well-oxygenated shallow waters where they graze on periphyton and detritus. Their distribution patterns often reflect substrate type, food availability, and calcium concentrations in the benthic environment. Quisque ullamcorper placerat ipsum. Cras nibh. Morbi vel justo vitae lacus tincidunt ultrices. Lorem ipsum dolor sit amet, consectetuer adipiscing elit.

Figure 3.8: Gastropoda

Amphipoda (F = 10.27***) was most abundant in Cluster 0 (1.728), moderate in Cluster 2 (1.227), and nearly absent in Cluster 1 (0.150). Scuds are sensitive indicators preferring clean, well-oxygenated habitats with structured substrates. They are important prey items for fish and serve as shredders and detritivores in the benthic food web, processing coarse particulate organic matter. Fusce mauris. Vestibulum luctus nibh at lectus. Sed bibendum, nulla a faucibus semper, leo velit ultricies tellus, ac venenatis arcu wisi vel nisl. Vestibulum diam.

Figure 3.9: Amphipoda

Nematoda (F = 5.71**) decreased progressively from Cluster 0 (3.478) to Cluster 1 (2.468) to Cluster 2 (1.283). Roundworms show distinct habitat preferences based on sediment characteristics and organic matter content. They occupy multiple trophic levels as bacterivores, fungivores, and predators, making them important components of the meiofaunal community in aquatic sediments. Suspendisse

Figure 3.10: Nematoda

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Chironomidae (F = 4.90*) showed lowest abundance in Cluster 0 (3.365) compared to Clusters 1 (4.689) and 2 (4.471). Non-biting midge larvae occupy a wide range of habitats with varying pollution tolerances. Different chironomid subfamilies and genera show distinct environmental preferences, making them valuable bioindicators when identified to lower taxonomic levels. Sed commodo posuere pede. Mauris ut est. Ut quis purus. Sed ac odio.

Figure 3.11: Chironomidae

Chapter 4

Environment Driven Taxa Clusters

Table 4.1: RDA Axes Summary

Axis	Eigenvalue	Expl. (%)	Cumul. (%)	F-stat	p-value
RDA1	0.0295	44.53	44.53	5.789	0.001**
RDA2	0.0155	23.42	67.95	3.044	0.151
RDA3	0.0119	17.98	85.93	2.338	0.483
RDA4	0.0047	7.15	93.08	0.929	1.000
RDA5	0.0026	3.98	97.06	0.517	1.000
RDA6	0.0020	2.94	100.00	0.383	1.000

Note: Permutations = 999; $R^2 = 0.2167$; Adjusted $R^2 = 0.1167$; Global test p-value = 0.0010**

Table 4.2: RDA Environmental Terms

Variable	Δ Inertia	F-stat	p-value	RDA1	RDA2
MPS (Phi)	1.204	4.459	0.001**	−0.900	0.088
Velocity (m/sec)	0.624	2.310	0.016*	0.361	0.165
Depth (m)	0.478	1.769	0.065	0.456	−0.442
Temperature (°C)	0.424	1.572	0.112	−0.046	0.699
DO Bottom (mg/L)	0.340	1.258	0.235	−0.108	−0.628
LOI (%)	0.185	0.684	0.747	0.136	0.228

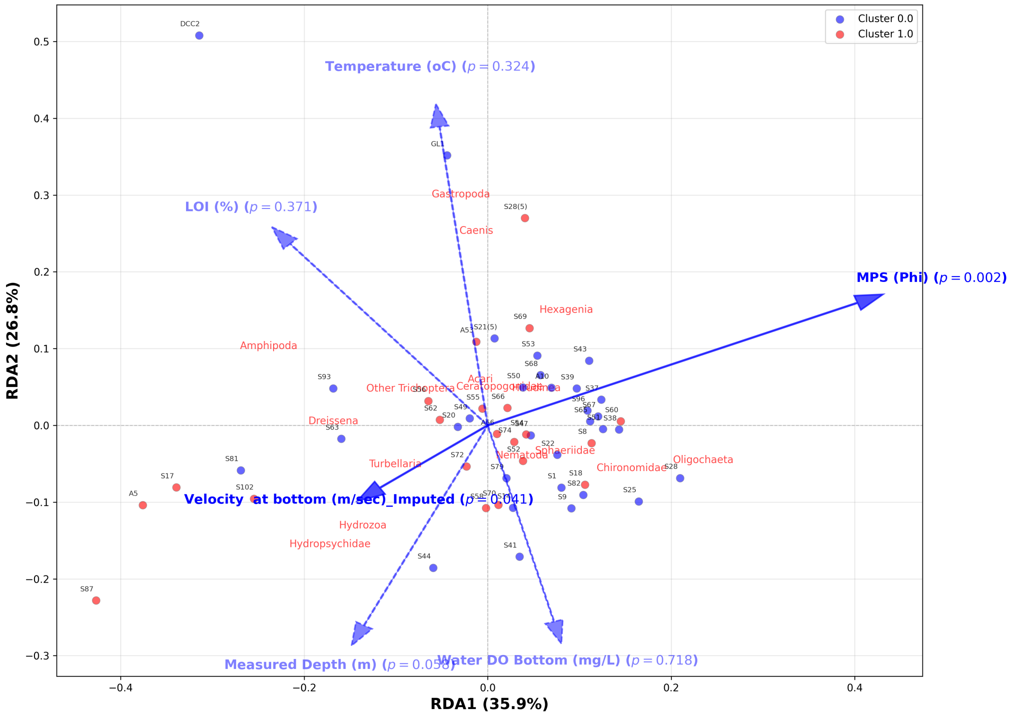

Figure 4.1: RDA Triplot

Table 4.3: LDA Variable Importance (Wilks' Lambda) and Discriminant Axes Summary

Variable	$\Delta\Lambda$	**F-stat**	**p-value**	**LD1**	**LD2**
MPS (Phi)	0.130	6.926	0.002**	−0.924	0.666
Temperature (°C)	0.119	6.336	0.004**	0.388	−0.667
Velocity (m/sec)	0.104	5.534	0.007**	−0.781	0.185
Depth (m)	0.087	4.617	0.015*	0.552	−0.010
DO Bottom (mg/L)	0.022	1.186	0.315	0.088	0.156
LOI (%)	0.000	0.024	0.976	−0.021	−0.013
Explained (%)	–	–	–	59.44	40.56
Cumulative (%)	–	–	–	59.44	100.00

Table 4.4: LDA Classification Confusion Matrix (Training Data)

	Pred. Cluster 0	**Pred. Cluster 1**	**Pred. Cluster 2**
True Cluster 0	12	5	1
True Cluster 1	3	25	0
True Cluster 2	1	3	4

Table 4.5: LDA Classification Report (Training Data)

	Precision	**Recall**	**F1-Score**	**Support**
Cluster 0	0.75	0.67	0.71	18
Cluster 1	0.76	0.89	0.82	28
Cluster 2	0.80	0.50	0.62	8
Accuracy	–	–	0.76	54
Macro avg	0.77	0.69	0.71	54
Weighted avg	0.76	0.76	0.75	54

Note: Overall Accuracy = 0.7593; Number of sites = 54

Table 4.6: LDA Monte Carlo Cross-Validation Aggregate Confusion Matrix (Test Size: 20.0% per 1000 iteration)

	Pred. Cluster 0	**Pred. Cluster 1**	**Pred. Cluster 2**
True Cluster 0	2178	1486	336
True Cluster 1	944	4901	155
True Cluster 2	237	389	374

Note: Combined results from 1,000 cross-validation iterations

Table 4.7: LDA Monte Carlo Cross-Validation Results Summary (Test Size: 20.0% per 1000 iteration)

Average Classification Performance				
Class	**Precision**	**Recall**	**F1-Score**	**Support**
Cluster 0	0.645 ± 0.275	0.544 ± 0.278	0.560 ± 0.236	18
Cluster 1	0.743 ± 0.134	0.817 ± 0.155	0.766 ± 0.108	28
Cluster 2	0.267 ± 0.384	0.374 ± 0.484	0.300 ± 0.406	8
Weighted Avg	0.664 ± 0.157	0.678 ± 0.123	0.649 ± 0.136	54
Mean accuracy		0.6775 ± 0.1228		
Median accuracy		0.7273		

Note: Classification metrics shown as mean ± standard deviation across 1,000 CV iterations

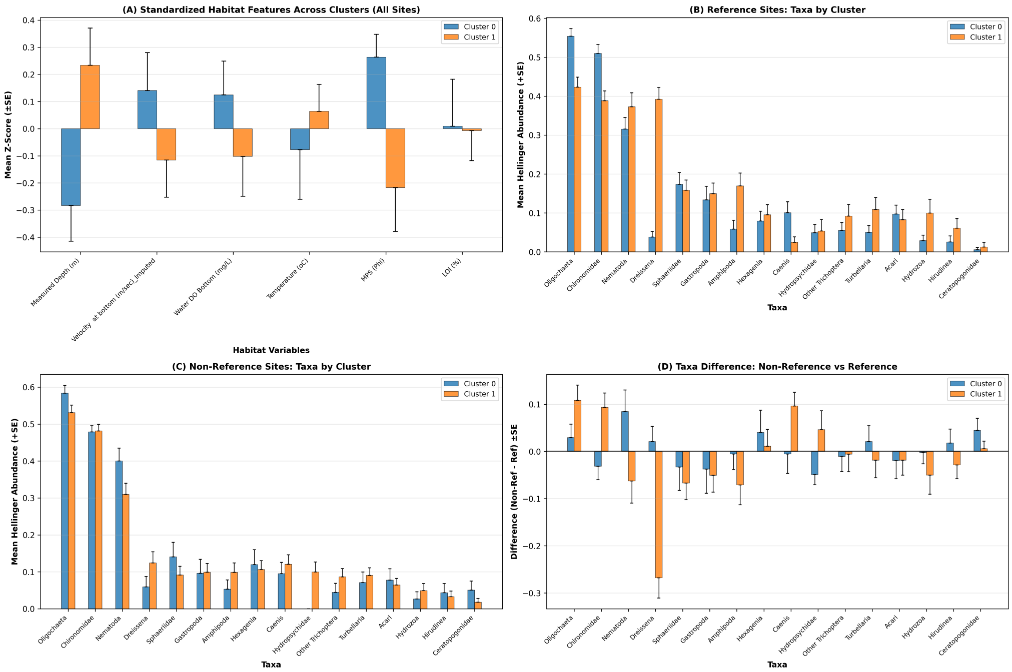

Figure 4.2: Cluster Comparison

Chapter 5

Community Composition Measures

Table 5.1: Significant Species PC vs Pollution Score Regressions

Cluster	**PC**	**Var.(%)**	R^2	$\bar{R}^2$	**Coef.**	t**-stat**	n
0	PC2	13.42	0.4223	0.4016	-0.0594	-4.524^{***}	30
2	PC1	35.16	0.5039	0.4625	-0.0907	-3.491^{**}	14

Note: Only significant regressions shown ($p < 0.01$). Significance: $^{**}p < 0.01$; $^{***}p < 0.001$

Table 5.2: Significant Pollution PC vs Species PC Regressions ($p < 0.05$)

Cluster	**Poll. PC**	**Spec. PC**	**Var.(%)**	R^2	**Slope**	n
2	PC2	PC1	35.16	0.4321	-0.151^{*}	14
0	PC3	PC2	13.42	0.2800	-0.159^{**}	30
0	PC4	PC5	8.46	0.2600	-0.097^{**}	30
0	PC2	PC2	13.42	0.2063	-0.096^{*}	30
0	PC2	PC4	9.81	0.1594	0.072^{*}	30
1	PC2	PC1	19.35	0.1172	0.085^{**}	60
1	PC1	PC2	15.06	0.1025	-0.057^{*}	60
1	PC2	PC6	6.40	0.0803	-0.041^{*}	60

Note: Var.(%) = variance explained by the Species PC.
Poll. PC = Pollution Principal Component; Spec. PC = Species Principal Component.
Significance: $^{*}p < 0.05$; $^{**}p < 0.01$

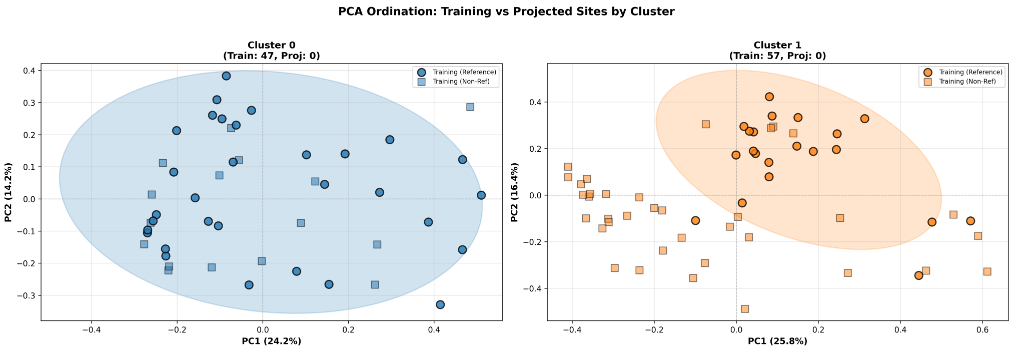

Figure 5.1: Ordination

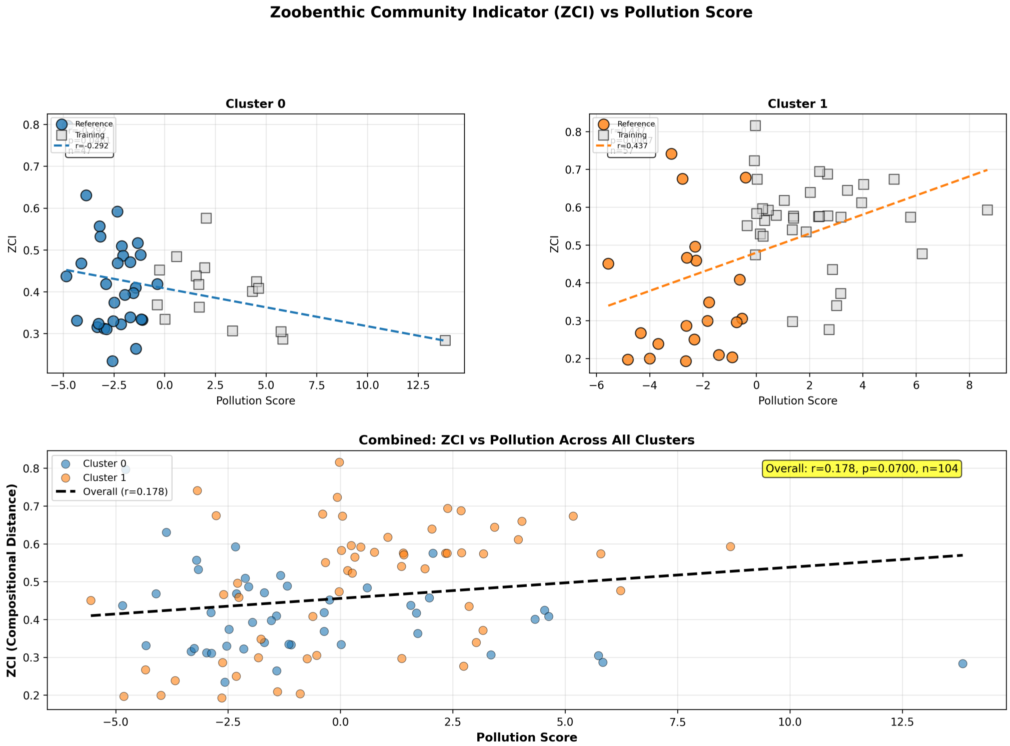

Figure 5.2: ZCI vs Pollution

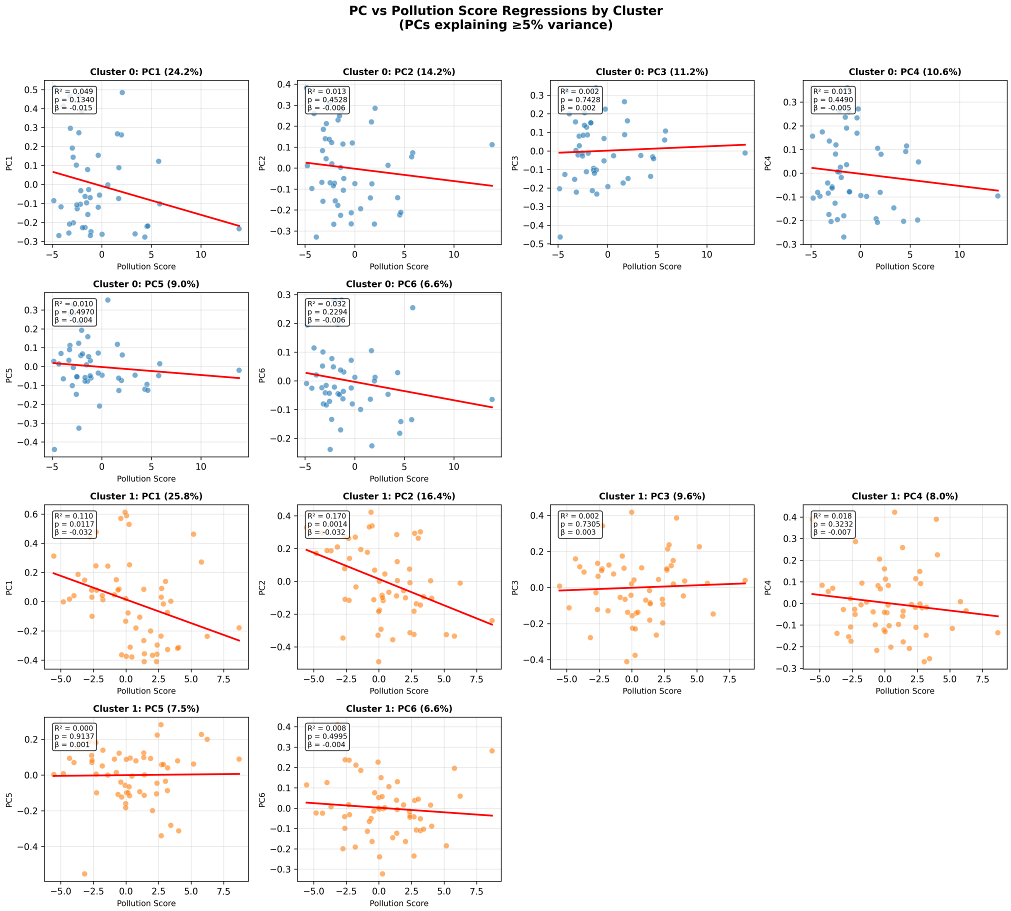

Figure 5.3: PC Regression

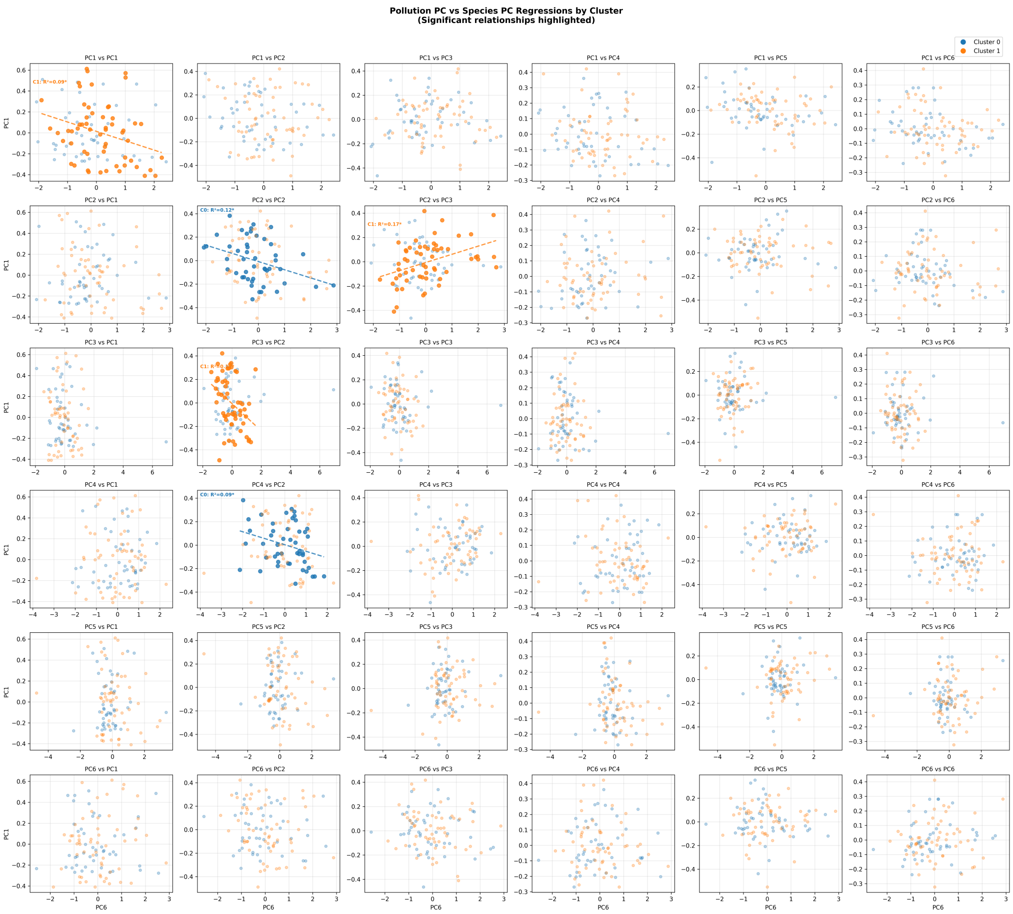

Figure 5.4: Pollution vs Species PC